Lumilens

Embedded Firmware Engineer

Firmware · Embedded Systems · Networking

What if your first job put you at the heart of the AI infrastructure revolution?

AI is moving incredibly fast.
And the infrastructure connecting it has to move even faster.

At Lumilens, we are building the connectivity platform for AI, the next-generation photonic interconnects designed to massively scale the AI infrastructure of tomorrow. We recently emerged from stealth with $900M+ in funding, and a valuation of $5.5 billion. At Lumilens we are building the critical photonics infrastructure that powers tomorrow's AI supercomputing. From chip-to-chip optical interconnects to scalable photonic engines, Lumilens is unlocking a new era of computing, faster, cooler, and massively more efficient.

$900M+	$5.5B
In funding, recently out of stealth	Company valuation

We are led by veterans who've built and scaled some of the most transformative technologies in the industry. This isn't incremental innovation, it's a ground-floor opportunity to rethink the optical layer from the silicon up. You'll work alongside a team of world-class engineers solving some of the hardest challenges in optics, systems, and scale. Every line of code, every design decision, every breakthrough you help deliver will shape the infrastructure of tomorrow.

But we're only getting started.

WHO THIS IS FOR

Early Talent at Lumilens is for people at the beginning of their careers who don't want to wait years to work on meaningful engineering problems. You will work alongside deeply experienced engineers on technology being designed, tested and built for real-world AI infrastructure.

No sidelines. / No waiting for "someday." / Come curious. Learn fast. Build for real.

01 ABOUT THE ROLE

As an Embedded Firmware Engineer at Lumilens, you will build the firmware layer that brings our networking and optical semiconductor products to life. Firmware is what configures the device, keeps the optical link running, and reports what is happening inside the silicon, so the code you write sits directly between our hardware and the customer networks it serves.

You will work primarily in C on ARM-based controllers, alongside hardware, DSP and software engineers, and use Python and shell scripting to automate the test and build environments around your work. It is a hands-on role with real ownership: you will take features from design review through to delivery, and see them ship inside infrastructure that powers tomorrow's AI supercomputing.

02 PRIMARY DUTIES & RESPONSIBILITIES

- Independently drive firmware design, development, and feature delivery for high-performance networking and optical semiconductor products
- Develop and maintain device drivers, real-time operating systems (RTOS) integration, and communication protocol implementations
- Design, code, test, and debug embedded firmware primarily in C for ARM-based controllers
- Collaborate closely with cross-functional teams including hardware, DSP, and software engineers
- Manage, configure, and optimize build environments and makefiles
- Contribute to design reviews, feature definition, and product delivery schedules

03 REQUIRED SKILLS & QUALIFICATIONS

- Bachelor's or Master’s degree in Computer Science, Electronics, Telecommunications, or related field.
- Strong proficiency in Python for test automation.
- Good understanding of Layer 1 and Layer 2 networking protocols.
- Experience with optical testing, network switches, and traffic generators like IXIA.
- Ability to debug issues in complex networking environments.
- Experience with Jenkins or ArgoCD other CI/CD tools.
- Familiarity with Shell scripting for automation and environment setup.
- Exposure to C# for UI validation or tool development is a plus.

04 WHAT WE OFFER

- ✓ Competitive salary commensurate with experience
- ✓ Comprehensive benefits package, including health insurance
- ✓ Professional development and certification opportunities
- ✓ Access to cutting-edge optical/photonics test environments and tools
- ✓ A collaborative, cross-functional team dedicated to solving some of the hardest challenges in photonics

Come curious. Learn fast. Build for real.

Equal opportunity employer: All qualified applicants will receive consideration without regard to race, color, religion, gender, identity, orientation, veteran status, disability, or any other legally protected status